

ARFOM-DB: PROBLEM DESCRIPTION & SYSTEM SCOPE

Airline Reservation and Flight Operations Management System

Course:	Database Management Systems (DBMS) - Presentation-I
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Section / Serial:	AIML Whales - Serial #18
Evaluation Stage:	Review 1 & Scope Definition

1. Executive Summary & Problem Statement

Modern airline operators manage complex, highly concurrent transactional environments spanning flight scheduling, fleet allocation, seat inventory, customer reservations, electronic ticketing, baggage handling, and financial settlements.

Traditional spreadsheet-driven and legacy file-based reservation mechanisms suffer from four critical failure modes:

- Double-Booking & Seat Contention: Inability to enforce strict multi-attribute uniqueness, leading to simultaneous booking of identical physical aircraft seats.
- Data Redundancy & Anomalies: Redundant storage of flight routes, passenger identities, and aircraft capacities across disparate logs leads to update, insertion, and deletion anomalies.
- Lack of Referential Integrity: Orphaned baggage items, inconsistent cancellation refunds, and untracked itinerary changes.
- Weak Domain Constraints: Unchecked domain values (e.g., negative luggage weights, flights departing and arriving at identical hubs, or chronological departure/arrival inversions).

ARFOM-DB resolves these operational challenges by designing and implementing an enterprise-grade, normalized relational database management system with rigorous integrity constraints, zero-loss 3NF decomposition, and high-performance transactional query interfaces.

2. Project Objectives

- Comprehensive Relational Modeling: Engineer an end-to-end database architecture encompassing 10+ core entities covering passenger lifecycles from booking to baggage claim.
- Strict Normalization (3NF/BCNF): Eliminate data redundancy and eliminate anomalies by decomposing Many-to-Many relationships via dedicated junction entities.
- Declarative Constraint Architecture: Enforce physical, temporal, and financial invariants at the storage engine level (PRIMARY KEY, FOREIGN KEY, CHECK, and UNIQUE).
- High-Concurrency Operational Performance: Support sub-millisecond multi-table analytical joins and real-time flight manifest projections.
- Interactive Management UI: Provide an intuitive, real-time web portal supporting CRUD operations with live database state reflection.

3. Scope of the System

- Fleet & Flight Infrastructure: Aircraft topology (airframe models, cabin seat classes: Economy, Business, First), airport registries, route networks with geographic distance validation, and operational flight schedules.
- Customer Reservation & Commercial Ticketing: Passenger demographic registration, multi-passenger booking orders (PNRs), individual flight leg ticket issuance, and double-booking prevention.
- Settlement & Audit Ledger: Multi-mode payment tracking (Credit Card, Debit Card, UPI, Net Banking), transaction verification, and cancellation fee/refund processing.
- Departure Control & Logistics: Airport check-in management, electronic boarding pass generation, baggage allowance tracking, and excess weight surcharge accounting.